

Unit 01: Number System

CONTENTS

Objective

Introduction

1.1 Types of Numbers

13. Composite numbers:

1.2 Multiplication

1.3 Distributive Laws

1.4 Division

1.5 Tests of Divisibility

1.6 Squares

1.7 Methods of Finding H.C.F.

1.8 Least Common Multiple

Summary

Keywords

Self Assessment

Review Questions

Further Readings

Objective

The key concepts explained in this unit, including area a learner is expected to learn and master after going through the unit are: -

- Understand what are different types of numbers
- Understand Learn Multiplication short cuts.
- Get Distributive Laws.
- Understand Division laws.
- Understand Tests of Divisibility
- Understand Methods of Finding L.C.M.
- Understand Methods of Finding H.C.F.

Introduction

Numbers are an integral part of our everyday lives, right from the number of hours we sleep at night to the number of rounds we run around the racing track and much more. In math, numbers can be even and odd numbers, prime and composite numbers, decimals, fractions, rational and irrational numbers, natural numbers, integers, real numbers, rational numbers, irrational numbers, and whole numbers. In this chapter, we'll get an introduction to the different types of numbers and to all the concepts related to it. Numbers form the basis of mathematics. We should befriend numbers in order to understand math. Numbers are of various kinds. We have a long list that includes ordinal numbers, consecutive numbers, odd numbers, even numbers, natural numbers, whole numbers, integers, real numbers, rational numbers, irrational numbers, and complex numbers.

Along with numbers, we come across the interesting world of factors and multiples. This world includes prime numbers, composite numbers, co-prime numbers, perfect numbers (yes, numbers could be perfect!) HCF, LCM, and prime factorization.

1.1 **Types of Numbers**

1. Natural Numbers: Counting numbers 1, 2, 3, 4, 5, are called natural numbers.
2. Whole Numbers: All counting numbers together with zero form the set of whole numbers.

Thus,

(i) 0 is the only whole number which is not a natural number.

(ii) Every natural number is a whole number.

3. Integers: All natural numbers, 0 and negatives of counting numbers i.e.,

$\{..., -3, -2, -1, 0, 1, 2, 3, \dots\}$ together form the set of integers.

(i) Positive Integers: $\{1, 2, 3, 4, \dots\}$ is the set of all positive integers.

(ii) Negative Integers: $\{-1, -2, -3, \dots\}$ is the set of all negative integers.

(iii) Non-Positive and Non-Negative Integers: 0 is neither positive nor negative. So, $\{0, 1, 2, 3, \dots\}$ represents the set of non-negative integers, while $\{0, -1, -2, -3, \dots\}$ represents the set of non-positive integers.

4. Even Numbers: A number divisible by 2 is called an even number, e.g., 2, 4, 6, 8, 10, etc.

5. Odd Numbers: A number not divisible by 2 is called an odd number. e.g., 1, 3, 5, 7, 9, 11, etc.

6. Prime Numbers: A number greater than 1 is called a prime number, if it has exactly two factors, namely 1 and the number itself.

Prime numbers upto 100 are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

7. Composite Numbers: Numbers greater than 1 which are not prime, are known as composite numbers, e.g., 4, 6, 8, 9, 10, 12.

Note: (i) 1 is neither prime nor composite.

(ii) 2 is the only even number which is prime.

(iii) There are 25 prime numbers between 1 and 100.

8. Co-primes: Two numbers a and b are said to be co-primes, if their H.C.F. is 1. e.g.,

(2, 3), (4, 5), (7, 9), (8, 11), etc. are co-primes.

8. Rational Numbers

The numbers of the form p

q, where p and q are integers

and $q \neq 0$, are known as rational numbers,

The set of all rational numbers is denoted by Q.

That is, $Q = \{x : x = \frac{p}{q}, p, q \in I, q \neq 0\}$

Since every natural number 'a' can be written as $\frac{a}{1}$, every natural number is a rational number. Since 0 can be written as $\frac{0}{1}$ and every non-zero integer 'a' can be written as $\frac{a}{1}$ every integer is a rational number.

Unit 01: Number System

Every rational number has a peculiar characteristic that when expressed in decimal form is expressible either in terminating decimals or in non-terminating repeating decimals.

For example, $\frac{1}{5} = 0.2$, $\frac{1}{3} = 0.333\dots$, $\frac{22}{7} = 3.1428714287\dots$, $\frac{8}{44} = 0.181818\dots$, etc.

The recurring decimals have been given a short notation as

$$0.333\dots = 0.\underline{3}$$

$$4.1555\dots = 4.0\underline{5}$$

$$0.323232\dots = 0.\underline{32}$$

9. Irrational numbers: Those numbers which when expressed in decimal form are neither terminating nor repeating decimals are known as irrational numbers, e.g., $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, π , etc.

Note that the exact value of π is not $\frac{22}{7}$. $\frac{22}{7}$ is rational while π is irrational number. $\frac{22}{7}$ is approximate value of π . Similarly, 3.14 is not an exact value of it.

10. Real numbers: The rational and irrational numbers combined together to form real numbers, e.g., $\frac{13}{21}$, $\frac{2}{5}$, $-\frac{3}{7}$, $\sqrt{3}$, $4 + \sqrt{2}$ etc. are real numbers. The set of all real numbers is denoted by R

Note that the sum, difference or product of a rational and irrational number is irrational, e.g., $3 + \sqrt{2}$, $4 - \sqrt{3}$, $\frac{2}{5} - \sqrt{5}$, $\sqrt[4]{3}$, $-\sqrt[3]{5}$ are all irrational.

11. Even numbers: All those numbers which are exactly divisible by 2 are called even numbers, e.g., 2, 6, 8, 10, etc., are even numbers.

12. Odd numbers: All those numbers which are not exactly divisible by 2 are called odd numbers, e.g., 1, 3, 5, 7, etc., are odd numbers.

13. Composite numbers: Natural numbers greater than 1 which are not prime, are known as composite numbers. For example, each of the numbers 4, 6, 8, 9, 12, etc., are composite numbers.

Note that the numbers which are relatively prime need not necessarily be prime numbers, e.g., 16 and 17 are relatively prime, although 16 is not a prime number.

1.2 Multiplication

1. Multiplication By Distributive Law :

(i) $a \times (b + c) = a \times b + a \times c$ (ii) $a \times (b - c) = a \times b - a \times c$.



Example 1.(i) $567958 \times 99999 = 567958 \times (100000 - 1)$

$$= 567958 \times 100000 - 567958 \times 1 = (56795800000 - 567958) = 56795232042. \text{ (ii) } 978 \times 184 + 978 \times 816 = 978 \times (184 + 816) = 978 \times 1000 = 978000.$$

2. Multiplication of a Number By 5^n : Put n zeros to the right of the multiplicand and divide the number so formed by 2^n .

$$\text{Ex. } 975436 \times 625 = 975436 \times 5^4 = 9754360000 = 609647600.$$

Multiplication of a given number by 9, 99, 999, etc., that is by $10^n - 1$ Method: Put as many zeros to the right of the multiplicand as there are nines in the multiplier and from the result subtract the multiplicand and get the answer.



Example 2: Multiply:

(a) 3893 by 99

$$3893 \times 99 = 389300 - 3893 = 385407$$

(b) 4327 by 999

$$4327 \times 999 = 4327000 - 4327 = 4322673$$

(c) 5863 by 99

$$5863 \times 9999 = 58630000 - 5863 = 58624137$$

Analytical Skills-I

Multiplication of a given number by 11, 101, 1001, etc., that is, by $10^n + 1$.

Method: Place n zeros to the right of the multiplicand and then add the multiplicand to the number so obtained.



Example 3: Multiply

$$4782 \times 11$$

$$\text{Solution: (a) } 4782 \times 11 = 47820 + 4782 = 52602$$

$$\text{b) } 9836 \times 101$$

$$\text{Solution :- } 9836 \times 101 = 983600 + 9836 = 993436$$

$$\text{c) } 6538 \times 1001$$

$$\text{Solution : - } 6538 \times 1001 = 6538000 + 6538 = 6544538$$

Multiplication of a given number by 15, 25, 35, etc. Method: Double the multiplier and then multiply the multiplicand by this new number and finally divide the product by 2



Example 4: Multiply

$$7054 \times 15$$

$$= \frac{1}{2}(7054 \times 30)$$

$$= \frac{1}{2}(211620)$$

$$= 105810$$

$$\text{b) } 3897 \times 25$$

$$= \frac{1}{2}(3897 \times 50)$$

$$= \frac{1}{2}(194850)$$

$$= 97425$$

$$\text{c) } 4536 \times 35$$

$$= \frac{1}{2}(4536 \times 70)$$

$$= \frac{1}{2}(319410)$$

$$= 159705$$



Example 5: Multiply

$$3982 \times 2$$

$$= \frac{39820}{2}$$

$$= 19910$$

$$\text{b) } 4739 \times 25$$

$$= \frac{473900}{2^2}$$

$$= 118475$$

$$\text{c) } 7894 \times 125$$

$$= \frac{7894000}{2^3}$$

$$= \frac{7894000}{8}$$

$$\text{d) } 4863 \times 625$$

$$= \frac{48630000}{2^4}$$

$$= \frac{48630000}{16}$$

$$= 3039375$$

1.3 Distributive Laws

For any three numbers a , b, c we have

a) $a \times b + a \times c = a \times (b + c)$

b) $A \times b - a \times c = a \times (b - c)$



Example 6: $438 \times 637 + 438 \times 367 = ?$

Solution: $- 438 \times 637 + 438 \times 367 = 438 \times (637 + 367) = 438 \times 1000 = 438000$



Example 7: $674 \times 832 - 674 \times 632 = ?$

Solution:- $674 \times 832 - 674 \times 632$

$$= 674 \times (832 - 632)$$

$$= 674 \times 200$$

$$= 134800$$

1.4 Division

Division is repeated subtraction. For example, when we divide 63289 by 43, it means 43 can be repeatedly subtracted 1471 times from 63289 and they remainder 36 is left.

$$\begin{array}{r}
 \text{Divisor} \rightarrow 43 \overline{) 63289} \quad \begin{array}{l} 1471 \leftarrow \text{Quotient} \\ \leftarrow \text{Dividend} \end{array} \\
 \underline{43} \\
 202 \\
 \underline{172} \\
 308 \\
 \underline{301} \\
 79 \\
 \underline{43} \\
 36 \leftarrow \text{Remainder}
 \end{array}$$

Dividend = (Divisor $\times$ quotient) + remainder

$$\text{Divisor} = \frac{\text{dividend} - \text{remainder}}{\text{Quotient}}$$



Example 8: On dividing 7865321 by a certain number, the quotient is 33612 and the remainder is 113. Find the divisor.

Solution: Divisor = Dividend - Remainder / Quotient

$$= \frac{7865321 - 113}{33612}$$

$$= \frac{7865208}{33612}$$

$$= 234$$



Example 9: A number when divided by 315 leaves remainder 46 and the value of quotient is 7. Find the number.

Solution: Number = (Divisor $\times$ Quotient) + Remainder = $(315 \times 7) + 46 = 2205 + 46 = 2251$

Example 10: Find the least number of 5 digits which is exactly divisible by 632.

Solution: The least number of 5 digits is 10000. Dividing this number by 632, the remainder is 520. So, the required number = $10000 + (632 + 520) = 10112$

$$\begin{array}{r} 15 \\ 632 \overline{) 10000} \\ \underline{632} \\ 3680 \\ \underline{3160} \\ 520 \end{array}$$



Example 11: Find the greatest number of 5 digits which is exactly divisible by 463.

Solution: The greatest number of 5 digits is 99999. Dividing this number by 463, the remainder is 454. So, the required number = $99999 - 454 = 99545$

$$\begin{array}{r} 215 \\ 463 \overline{) 99999} \\ \underline{926} \\ 739 \\ \underline{463} \\ 2769 \\ \underline{2315} \\ 454 \end{array}$$



Example 12: Find the number nearest to 13700 which is exactly divisible by 235.

Solution: On dividing the number 13700 by 235, the remainder is 70. Therefore, the nearest number to 13700, which is exactly divisible by 235 = $13700 - 70 = 13630$.

$$\begin{array}{r} 58 \\ 235 \overline{) 13700} \\ \underline{1175} \\ 1950 \\ \underline{1880} \\ 70 \end{array}$$

1.5 Tests of Divisibility

1. Divisibility by 2:- A number is divisible by 2 if the unit's digit is zero or divisible by 2. For example, 4, 12, 30, 18, 102, etc., are all divisible by 2.

2. Divisibility by 3 :- A number is divisible by 3 if the sum of digits in the number is divisible by 3.

For example, the number 3792 is divisible by 3 since $3 + 7 + 9 + 2 = 21$, which is divisible by 3.

3. Divisibility by 4:- A number is divisible by 4 if the number formed by the last two digits (ten's digit and unit's digit) is divisible by 4 or are both zero. For example, the number 2616 is divisible by 4 since 16 is divisible by 4.

4. Divisibility by 5:- A number is divisible by 5 if the unit's digit in the number is 0 or 5. For example, 13520, 7805, 640, 745, etc., are all divisible by 5.

5. Divisibility by 6 A number is divisible by 6 if the number is even and sum of its digits is divisible by 3. For example, the number 4518 is divisible by 6 since it is even and sum of its digits $4 + 5 + 1 + 8 = 18$ is divisible by 3.

6. Divisibility by 7:- The unit digit of the given number is doubled and then it is subtracted from the number obtained after omitting the unit digit. If the remainder is divisible by 7, then the given number is also divisible by 7. For example, consider the number 448. On doubling the unit digit 8 of 448 we get 16. Then, $44 - 16 = 28$.

Since 28 is divisible by 7, 448 is divisible by 7.

7. Divisibility by 8:- A number is divisible by 8, if the number formed by the last 3 digits is divisible by 8.

Unit 01: Number System

For example, the number 41784 is divisible by 8 as the number formed by last three digits, i.e., 784 is divisible by 8.

8. Divisibility by 9:- A number is divisible by 9 if the sum of its digits is divisible by 9. For example, the number 19044 is divisible by 9 as the sum of its digits $1 + 9 + 0 + 4 + 4 = 18$ is divisible by 9.

9. Divisibility by 10:- A number is divisible by 10, if it ends in zero.

For example, the last digit of 580 is zero, therefore, 580 is divisible by 10.

10. Divisibility by 11:- A number is divisible by 11 if the difference of the sum of the digits at odd places and sum of the digits at even places is either zero or divisible by 11.

For example, in the number 38797, the sum of the digits at odd places is $3 + 7 + 7 = 17$ and the sum of the digits at even places is $8 + 9 = 17$. The difference is $17 - 17 = 0$, so the number is divisible by 11.

11. Divisibility by 12 :- A number is divisible by 12 if it is divisible by 3 and 4.

12. Divisibility by 18:- An even number satisfying the divisibility test of 9 is divisible by 18.

13. Divisibility by 25 A number is divisible by 25 if the number formed by the last two digits is divisible by 25 or the last two digits are zero. For example, the number 13675 is divisible by 25 as the number formed by the last two digits is 75 which is divisible by 25

14. Divisibility by 88 A number is divisible by 88 if it is divisible by 11 and 8.

15. Divisibility by 125 A number is divisible by 125 if the number formed by the last three digits is divisible by 125 or the last three digits are zero. For example, the number 5250 is divisible by 125 as 250 is divisible by 125.

1.6 Squares

To square any number ending with 5.

Method: $(A5)^2 = A(A + 1)/25$



Example 13:

(a) $(25)^2 = 2(2 + 1)/25 = 6/25 = 625$

(b) $(45)^2 = 4(4 + 1)/25 = 20/25 = 2025$

(c) $(85)^2 = 8(8 + 1)/25 = 72/25 = 7225$

To square a number in which every digit is one. Method: Count the number of digits in the given number and start writing numbers in ascending order from one to this number and then in descending order up to one.



Example 14:

(a) $11^2 = 121$

(b) $111^2 = 12321$

(c) $1111^2 = 1234321$

(d) $222^2 = 2^2 (111)^2 = 4(12321) = 49284$

(e) $3333^2 = 3^2 (1111)^2 = 9(1234321) = 11108889$

To square a number which is nearer to $10x$. Method: Use the formula: $x^2 = (x^2 - y^2) + y^2 = (x + y)(x - y) + y^2$



Example 15:

(a) $(97)^2 = (97 + 3)(97 - 3) + 3^2 = 9400 + 9 = 9409$

(b) $(102)^2 = (102 - 2)(102 + 2) + 2^2 = 10400 + 4 = 10404$

(c) $(994)^2 = (994 + 6)(994 - 6) + 6^2 = 988000 + 36 = 988036$

(d) $(1005)^2 = (1005 - 5)(1005 + 5) + 5^2 = 1010000 + 25 = 1010025$

Common Factor

If a number a divides another number b exactly, we say that a is a factor of b . In this case, b is called a multiple of a . A common factor of two or more numbers is a number which divides each of them exactly. For example, 4 is a common factor of 8 and 12.

Highest Common Factor

Highest common factor of two or more numbers is the greatest number that divides each one of them

Exactly For example, 6 is the highest common factor of 12, 18 and 24. Highest Common Factor is also called Greatest Common Divisor or Greatest Common Measure. Symbolically, these can be written as H.C.F. or G.C.D. or G.C.M., respectively.

1.7 Methods of Finding H.C.F.

I. Method of Prime Factors

Step1 Express each one of the given numbers as the product of prime factors. [A number is said to be a prime number if it is exactly divisible by 1 and itself, but not by any other number, e.g., 2, 3, 5, 7, etc. are prime numbers]

Step 2 Choose common factors.

Step 3 Find the product of these common factors. This is the required H.C.F. of given numbers.



Example 16: Find the H.C.F. of 70 and 90.

$$\text{Solution: } 70 = 2 \times 5 \times 7$$

$$90 = 2 \times 5 \times 9$$

Common factors are 2 and 5.

$$\therefore \text{H.C.F.} = 2 \times 5 = 10.$$



Example 17: Find the H.C.F. of 3332, 3724 and 4508.

$$\text{Solution: } 3332 = 2 \times 2 \times 7 \times 7 \times 17$$

$$3724 = 2 \times 2 \times 7 \times 7 \times 19$$

$$4508 = 2 \times 2 \times 7 \times 7 \times 23$$

$$\therefore \text{H.C.F.} = 2 \times 2 \times 7 \times 7 = 196.$$



Example 18: Find the H.C.F. of 360 and 132.

$$\text{Solution: } 360 = 2^3 \times 3^2 \times 5$$

$$132 = 2^2 \times 3^1 \times 11$$

$$\therefore \text{H.C.F.} = 2^2 \times 3^1 = 12.$$



Example 19: If $x = 2^3 \times 3^5 \times 5^9$ and $y = 2^5 \times 3^7 \times 5^{11}$, find H.C.F. of x and y .

Solution: The factors common to both x and y are $2^3 3^5$ and 5^9 .

$$\therefore \text{H.C.F.} = 2^3 \times 3^5 \times 5^9.$$

II. Method of Division

A. For two numbers:

Step 1 Greater number is divided by the smaller one.

Step 2 Divisor of (1) is divided by its remainder.

Step 3 Divisor of (2) is divided by its remainder. This is continued until no remainder is left.

H.C.F. is the divisor of last step.



Example 20: Find the H.C.F. of 3556 and 3444.

$$\begin{array}{r}
 3444 \overline{) 3556} \quad 1 \\
 \underline{3444} \\
 112 \overline{) 3444} \quad 30 \\
 \underline{3360} \\
 84 \overline{) 112} \quad 1 \\
 \underline{84} \\
 28 \overline{) 84} \quad 3 \\
 \underline{84} \\
 \times
 \end{array}$$

$\therefore$ H.C.F. = 28.

A. For more than two numbers:

Step 1 Any two numbers are chosen and their H.C.F. is obtained.

Step 2 H.C.F. of H.C.F. (of (1)) and any other number is obtained.

Step 3 H.C.F. of H.C.F. (of (2)) and any other number (not chosen earlier) is obtained.

This process is continued until all numbers have been chosen. H.C.F. of last step is the required H.C.F.



Example 21: Find the H.C.F. of 13915, 9499 and 2553 by division method.

Solution:

$$\begin{array}{r}
 9499 \overline{) 13915} \quad 1 \\
 \underline{9499} \\
 4416 \overline{) 9499} \quad 2 \\
 \underline{8832} \\
 667 \overline{) 4416} \quad 6 \\
 \underline{4002} \\
 414 \overline{) 667} \quad 1 \\
 \underline{414} \\
 253 \overline{) 414} \quad 1 \\
 \underline{253} \\
 161 \overline{) 253} \quad 1 \\
 \underline{161} \\
 92 \overline{) 161} \quad 1 \\
 \underline{92} \\
 69 \overline{) 92} \quad 1 \\
 \underline{69} \\
 23 \overline{) 69} \quad 3 \\
 \underline{69} \\
 \times
 \end{array}$$

we will find the H.C.F. of 23 and 2553.

Now, in the next step,

$$\begin{array}{r} 23 \overline{) 2553} \quad 111 \\ \underline{23} \\ 25 \\ \underline{23} \\ 23 \\ \underline{23} \\ 0 \end{array}$$

Thus, H.C.F. of 13915, 9499 and 2553 = 23.

Example 22: Find the greatest possible length which can be used to measure exactly the lengths 7 m, 3 m 85 cm, 12 m 95 cm.

Solution: Required length

$$= (\text{H.C.F. of } 700, 385, 1295) \text{ cm} = 35 \text{ cm.}$$

Common multiple

A common multiple of two or more numbers is a number which is exactly divisible by each one of them. For example, 32 is a common multiple of 8 and 16.

$$8 \times 4 = 32$$

$$16 \times 2 = 32.$$

1.8 Least Common Multiple

The least common multiple of two or more given numbers is the least or lowest number which is exactly divisible by each of them.

For example, consider the two numbers 12 and 18.

Multiples of 12 are 12, 24, 36, 48, 60, 72, ...

Multiples of 18 are 18, 36, 54, 72, ...

Common multiples are 36, 72, ...

$\therefore$ Least common multiple, i.e., L.C.M. of 12 and 18 is 36.

Methods of Finding L.C.M.

Method of Prime Factors

Step 1 Resolve each given number into prime factors.

Step 2 Take out all factors with highest powers that occur in given numbers.

Step 3 Find the product of these factors. This product will be the L.C.M.



Example 23: Find the L.C.M. of 32, 48, 60 and 320.

$$\text{Solution: } 32 = 2^5 \times 1$$

$$48 = 2^4 \times 3$$

$$60 = 2^2 \times 3 \times 5$$

$$320 = 2^6 \times 5$$

$$\therefore \text{L.C.M.} = 2^6 \times 3 \times 5 = 960.$$

II. Method of Division

Step 1 The given numbers are written in a line separated by common.

Step 2 Divide by any one of the prime numbers 2, 3, 5, 7, 11, ... which will divide at least any two of the given numbers exactly. The quotients and the undivided numbers are written in a line below the first.

Step 3 Step 2 is repeated until a line of numbers (prime to each other) appears.

Unit 01: Number System

Step 4 Find the product of all divisors and numbers in the last line, which is the required L.C.M.



Example 24: Find the L.C.M. of 12, 15, 20 and 54.

Solution:

2	12, 15, 20, 54
2	6, 15, 10, 27
3	3, 15, 5, 27
5	1, 5, 5, 9
	1, 1, 1, 9

$$\text{L.C.M.} = 2 \times 2 \times 3 \times 5 \times 1 \times 1 \times 1 \times 9 = 540.$$

Notes

Before finding the L.C.M. or H.C.F., we must ensure that all quantities are expressed in the same unit

Short-cut methods

01 H.C.F. and L.C.M. of Decimals

Step 1 Make the same number of decimal places in all the given numbers by suffixing zero(s) if necessary.

Step 2 Find the H.C.F./L.C.M. of these numbers without decimal.

Step 3 Put the decimal point (in the H.C.F./L.C.M. of Step 2) leaving as many digits on its right as there are in each of the numbers.

Illustration 10: Find the L.C.M. of 1.2, 0.24 and 6. Solution: The given numbers can be written as 1.20, 0.24 and 6.00.

Now, ignoring the decimal we find the L.C.M. of 120, 24 and 600.

2	120, 24, 600
2	60, 12, 300
2	30, 6, 150
3	15, 3, 75
5	5, 1, 25
	1, 1, 5

$$\therefore \text{L.C.M.} = 2 \times 2 \times 2 \times 3 \times 5 \times 1 \times 5 = 600$$

Thus, the required L.C.M. = 6.00, i.e., 6.



Example 25: Find the H.C.F. of 6.16 and 13. Solution: The given numbers can be written as 6.16 and 13.00.

Now, ignoring the decimals we find the H.C.F. of 616 and 1300.

$$\begin{array}{r}
 616 \overline{) 1300} \quad (2 \\
 \underline{1232} \\
 68 \quad 616 (9 \\
 \underline{612} \\
 4 \quad 68 (17 \\
 \underline{68} \\
 \times
 \end{array}$$

$\therefore$ H.C.F. of 616 and 1300 is 4. Thus, the required H.C.F. = 0.04

02 L.C.M. and H.C.F. of Fractions

$$\text{L.C.M.} = \frac{\text{L.C.M. of the numbers in numerators}}{\text{H.C.F. of the numbers in denominators}}$$

$$\text{H.C.F.} = \frac{\text{H.C.F. of the numbers in numerators}}{\text{L.C.M. of the numbers in denominators}}$$

Example 26: Find the L.C.M. of $\frac{2}{5}$, $\frac{3}{10}$, and $\frac{6}{25}$

Solution: L.C.M. of numerators 2, 3 and 6 is 6.

H.C.F. of denominators 5, 10 and 25 is 5.

$$\therefore \text{Required L.C.M.} = \frac{\text{L.C.M. of Numerators}}{\text{H.C.F. of Denominators}} = \frac{6}{5}$$

Illustration 13: Find the H.C.F. of $\frac{4}{9}$, $\frac{10}{21}$ and $\frac{20}{63}$

Solution: H.C.F. of numerators 4, 10 and 20 is 2. L.C.M. of denominators 9, 21 and 63 is 63.

$$\therefore \text{Required H.C.F} = \frac{\text{H.C.F. of numerators}}{\text{L.C.M. of denominators}} = \frac{2}{63}$$

Notes

1. If the given set of numbers includes fractions as well as whole numbers, treat whole number too as fraction with 1 in its denominator.

2. The H.C.F. of a number of fractions is always a fraction, but the L.C.M. may be a fraction or an integer.

03 Product of two numbers = L.C.M. of the numbers $\times$ H.C.F. of the numbers



Example 27: The H.C.F. and the L.C.M. of any two numbers are 63 and 1260, respectively. If one of the two numbers is 315, find the other number.

Solution: The required number

$$= \frac{\text{L.C.M} \times \text{H.C.F}}{\text{First number}} = \frac{1260 \times 63}{315} = 252$$

04 To find the greatest number that will exactly divide x, y and z.

Required number = H.C.F. of x, y and z.



Example 28: Find the greatest number that will exactly divide 200 and 320. Solution: The required greatest number = H.C.F. of 200 and 320 = 40.

05 To find the greatest number that will divide x, y and z leaving remainders a, b and c, respectively. Required number = H.C.F. of (x - a), (y - b) and (z - c).

Illustration 16: Find the greatest number that will divide 148, 246 and 623 leaving remainders 4, 6 and 11, respectively.

Solution: The required greatest number = H.C.F. of (148 - 4), (246 - 6) and (623 - 11), i.e., H.C.F. of 144, 240 and 612 = 12.

06 To find the least number which is exactly divisible by x, y and z.

Required number = L.C.M. of x, y and z.



Example 29: What is the smallest number which is exactly divisible by 36, 45, 63 and 80?

Solution: The required smallest number = L.C.M. of 36, 45, 63 and 80 = 5040.

07 To find the least number which when divided by x, y and z leaves the remainders a, b and c, respectively. It is always observed that (x - a) = (y - b) = (z - c) = k (say)

$\therefore$ Required number = (L.C.M. of x, y and z) - k.



Example 30: Find the least number which when divided by 36, 48 and 64 leaves the remainders 25, 37 and 53, respectively.

Solution: Since, (36 - 25) = (48 - 37) = (64 - 53) = 11, therefore, the required smallest number = (L.C.M. of 36, 48 and 64) - 11 = 576 - 11 = 565.

08 To find the least number which when divided by x, y and z leaves the same remainder r in each case.

Required number = (L.C.M. of x, y and z) + r.



Example 31: Find the least number which when divided by 12, 16 and 18, will leave in each case a remainder 5.

Solution: The required smallest number = (L.C.M. of 12, 16 and 18) + 5 = 144 + 5 = 149.

09 To find the greatest number that will divide x, y and z leaving the same remainder in each case.

(a) When the value of remainder r is given: Required number = H.C.F. of $(x - r)$, $(y - r)$ and $(z - r)$.

(b) When the value of remainder is not given: Required number = H.C.F. of $|x - y|$, $|y - z|$ and $|z - x|$



Example 32: Find the greatest number which will divide 772 and 2778 so as to leave the remainder 5 in each case.

Solution: The required greatest number = H.C.F. of $(772 - 5)$ and $(2778 - 5)$ = H.C.F. of 767 and 2773 = 59.



Example 33: Find the greatest number which on dividing 152, 277 and 427 leaves equal remainder. Solution: The required greatest number = H.C.F. of $|x - y|$, $|y - z|$ and $|z - x|$ = H.C.F. of $|152 - 277|$, $|277 - 427|$ and $|427 - 152|$ = H.C.F. of 125, 150 and 275 = 25.

10 To find the n-digit greatest number which, when divided by x, y and z,

leaves no remainder (i.e., exactly divisible)

Step 1 L.C.M. of x, y and z = L

Step 2 $\frac{\text{n digit greatest number}}{\text{Remainder} = R}$

Step 3 Required number = n-digit greatest number - R

(b) leaves remainder K in each case Required number = (n-digit greatest number - R) + K.



Example 34: Find the greatest number of 4-digit number which, when divided by 12, 18, 21 and 28 leaves 3 as a remainder in each case.

Solution: L.C.M. of 12, 18, 21 and 28 = 252

$$\begin{array}{r} 252 \overline{) 9999} \quad 39 \\ \underline{9828} \\ 171 \end{array}$$

$\therefore$ The required number = $(9999 - 171) + 3 = 9931$.



Example 35: Find the greatest number of four digits which, when divided by 12, 15, 20 and 35 leaves no remainder.

Solution: L.C.M. of 12, 15, 20 and 35 = 420.

$$\begin{array}{r} 420 \overline{) 9999} \quad 23 \\ \underline{9660} \\ 339 \end{array}$$

$\therefore$ The required number = $9999 - 339 = 9663$.

11 To find the n-digit smallest number which when divided by x, y and z (a) leaves no remainder (i.e., exactly divisible)

Step 1 L.C.M. of x, y and z = L

Step 2 $\frac{\text{n digit smallest number}}{\text{Remainder} = R}$

Step 3 Required number = n-digit smallest number + (L - R).

(b) leaves remainder K in each case. Required number = n-digit smallest number + (L - R) + K.



Example 36: Find the least number of four digits which is divisible by 4, 6, 8 and 10.

Solution: L.C.M. of 4, 6, 8 and 10 = 120.

$$\begin{array}{r} 120 \overline{)1000} (8 \\ \underline{960} \\ 40 \end{array}$$

∴ The required number = $1000 + (120 - 40) = 1080$.



Example 37: Find the smallest 4-digit number, such that when divided by 12, 18, 21 and 28, it leaves remainder 3 in each case.

Solution: L.C.M. of 12, 18, 21 and 28 = 252.

$$\begin{array}{r} 252 \overline{)1000} (3 \\ \underline{756} \\ 244 \end{array}$$

∴ The required number = $1000 + (252 - 244) + 3 = 1011$.

Cyclicity of unit digit

Numbers are classified into three categories to find unit digit.

1. Digits 0,1,5,6
2. Digits 4,9
3. Digits 2,3,7,8

Digits 0,1,5,6

When we have these numbers (0,1,5,6) in the unit place, we get the same digit itself at the unit place when raised to any power, i.e. $0^n=0$, $1^n=1$, $5^n=5$, $6^n=6$. Let us apply this concept to the following questions.



Example 38: Find the Unit place digit of the following numbers:

1. 360^{244}

Answer= 0

2. 2974281^{307}

Answer=1

3. $4575^{400000666}$

Answer=5

4. $5687686^{265749375}$

Answer=6

Digits 4 & 9

Both these numbers have a cyclicity of only two different digits as their unit's digit.

In the case of 4 & 9

- If the Power of 4 is Even, the result will be 6
- If the Power of 4 is Odd, the result will be 4
- If the Power of 9 is Even, the result will be 1
- If the Power of 9 is Odd, the result will be 9.



Example: Find the Unit place digit of the following numbers:

1. $4568474^{26734258}$

Answer= 6.

2. 3456445767843

Answer= 4

3. $54857465789^{5768454}$

Answer= 1

4. $4576348567895627369^{765787}$

Answer= 9

Digits 2,3,7,8

For Digit 2

Unit 01: Number System

When we have number 2 in the unit place then follow the given steps to find the unit digit.

Step 1- Divide the last two digits of the power of a given number with 4

Step 2- You get the remainder n

Step 3- Since you have got n as a remainder, put it as the power of 2, i.e $(2)^n$

Step 4- Have a look at the table below and mark your answer.



Example 39: Find the Unit place digit of the following numbers:

1. 46572^{33}

Here, the unit place is 2 and power is 33. To solve follow the given steps

Step 1- Divide 33 by 4.

Step 2- You get remainder 1.

Step 3- Since you have got remainder 1, put it as a power of 2, i.e $(2)^1$

Step 4- Have a look at the table above, $(2)^1=2$.

Answer= 2.



Example 40: Find the Unit place digit of the following numbers:

1. 46573^{33}

Here, the unit place is 3 and power is 33. To solve follow the given steps

Step 1- Divide 33 by 4.

Step 2- You get remainder 1.

Step 3- Since you have got remainder 1, put it as a power of 3, i.e $(3)^1$

Step 4- Have a look at the table above, $(3)^1=3$.

Answer= 3.

For Digit 7



Example 41: Find the Unit place digit of the following numbers:

1. 46577^{18}

Here, the unit place is 7 and power is 18. To solve follow the given steps

Step 1- Divide 18 by 4.

Step 2- You get remainder 2.

Step 3- Since you have got remainder 2, put it as a power of 7, i.e $(7)^2$

Step 4- Have a look at the table above, $(7)^2=9$.

Answer= 9.

For Digit 8



Example 42: Find the Unit place digit of the following numbers:

1. 46578^{59}

Here unit place is 8 and power is 59. To solve follow the given steps

Step 1- Divide 59 by 4.

Step 2- You get remainder 3.

Step 3- Since you have got remainder 3, put it as a power of 8, i.e $(8)^3$

Step 4- Have a look at the table above, $(8)^3=2$.

Answer= 2

Summary

The key concepts learned from this unit are: -

- We have learnt what are different types of numbers
- We have learnt Multiplication short cuts.
- We have learnt Distributive Laws.
- We have learnt Division laws.
- We have learnt Tests of Divisibility
- We have learnt Methods of Finding L.C.M.
- We have learnt Methods of Finding H.C.F.

Keywords

- Multiplication.
- Distributive Laws.
- Division laws.
- Tests of Divisibility
- L.C.M.
- H.C.F.

Self Assessment

1. the unit's digit in the product $(2467)163 \times (341)72$
A. 1
B. 0
C. 7
D. 9
2. $387 \times 387 + 114 \times 114 + 2 \times 387 \times 114 =$
A. 251001
B. 251002
C. 251003
D. 251006
3. Which of the following statements is true?
A. 541326 is divisible by 3, 5967013 is not divisible by 3
B. 541326 is not divisible by 3, 5967013 is not divisible by 3
C. 541326 is divisible by 3, 5967013 is divisible by 3
D. 541326 is not divisible by 3, 5967013 is divisible by 3
4. Which of the following statements is true?
A. 67920594 is divisible by 4, 618703572 is not divisible by 4
B. 67920594 is not divisible by 4, 618703572 is not divisible by 4
C. 67920594 is divisible by 4, 618703572 is divisible by 4
D. 67920594 is not divisible by 4, 618703572 is divisible by 4
5. 4832718 is divisible by ?
A. 11
B. 5
C. 10
D. 25
6. Least number must be added to 3000 to obtain a number exactly divisible by 19 is?
A. 1
B. 2
C. 3
D. 4
7. The H.C.F. of 108, 288 and 360 is?
A. 24
B. 28
C. 36

D. 48

8. The H.C.F. of 513, 1134 and 1215 is?

- A. 13
- B. 27
- C. 81
- D. 124

9. The L.C.M. of 72, 108 and 2100 is?

- A. 38700
- B. 32500
- C. 37800
- D. 32

10. The L.C.M. of 16, 24, 36 and 54 is?

- A. 432
- B. 411
- C. 32
- D. 24

11. The H.C.F. and L.C.M. of 0.63, 1.05 and 2.1 is?

- A. H.C.F. of 0.63, 1.05 and 2.1 is 0.21, L.C.M. of 0.33, 1.05 and 2.1 is 5
- B. H.C.F. of 0.63, 1.05 and 2.1 is 0.21, L.C.M. of 0.93, 1.05 and 2.1 is 6.30
- C. H.C.F. of 0.63, 1.05 and 2.1 is 0.21, L.C.M. of 0.63, 1.05 and 2.1 is 6.30
- D. H.C.F. of 0.63, 1.05 and 2.1 is 0.21, L.C.M. of 0.63, 1.05 and 2.1 is 9.30

12. Two numbers are in the ratio of 15:11. If their H.C.F. is 13, then the numbers are?

- A. 195 and 200.
- B. 200 and 143.
- C. 195 and 145.
- D. 195 and 143.

13. The H.C.F. of two numbers is 11 and their L.C.M. is 693. If one of the

14. numbers is 77, then the other is?

- A. 95
- B. 99
- C. 120
- D. 200

15. $(994)^2 =$

- A. 1988036
- B. 998036
- C. 108036
- D. 988036

16. The Unit place digit of the number 46572^{33} is

- A. 2
- B. 3
- C. 4
- D. 5

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. C | 2. A | 3. A | 4. D | 5. A |
| 6. B | 7. C | 8. B | 9. C | 10. A |
| 11. D | 12. D | 13. B | 14. D | 15. A |

Review Questions

1. Find the greatest possible length which can be used to measure exactly the lengths 4 m 95 cm, 9 m and 16 m 65 cm.
2. Find the greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively.
3. Find the largest number which divides 62, 132 and 237 to leave the same remainder in each case
4. Find the least number exactly divisible by 12,15,20,27.
5. Find the least number which when divided by 6,7,8,9, and 12 leave the same remainder 1 each case.
6. The traffic lights at three different road crossings change after every 48 sec., 72 sec and 108 sec. respectively .If they all change simultaneously at 8:20:00hours,then at what time they again change simultaneously.
7. A number when divided by 315 leaves remainder 46 and the value of quotient is 7. Find the number.
8. Find the least number of 5 digits which is exactly divisible by 632.
9. Find the greatest number of 5 digits which is exactly divisible by 463.
10. Find the number nearest to 13700which is exactly divisible by 235.



Further Readings

1. Quantitative Aptitude For Competitive Examinations By Dr. R S Aggarwal, S Chand Publishing
2. A Modern Approach To Verbal & Non-Verbal Reasoning By Dr. R S Aggarwal, S Chand Publishing
3. Magical Book On Quicker Maths By M Tyra, Banking Service Chronicle
4. Analytical Reasoning By M.K. Pandey, Banking Service Chronicle